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Centre Number

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Student Number

2022 YEAR 12

Advanced Mathematics

Task 1 Validation Task

05/12/22

Q	Marks
1	/2
2	/2
3	/6
4	/2
5	/2
6	/5
7	/2
8	/6
9	/3
Total	/30

General

Instructions:

- Reading time – 5 minutes
- Working time – 35 minutes
- Write using blue or black pen
- NESA approved calculators may be used
- Show relevant mathematical reasoning and/or calculations
- No white-out may be used

Total Marks: 30

This question paper must not be removed from the examination room.

*This Validation task constitutes **60%** of the Assessment Task.*

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In Questions 1 – 9, your response should include relevant mathematical reasoning and/or calculations.

Question 1 (2 marks)

If $\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$, find:

(a) $\log_{10} 6$

1

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(b) $\log_{10} \frac{2}{9}$

1

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Question 2 (2 marks)

Solve the following using change of base. Write your answer to three significant figures.

2

$$5^x = 3$$

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Examination continues on next page

Question 3 (6 marks)

Solve for x if:

(a) $121^x = 11$

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(b) $5^{x+3} = 25$

1

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(c) $\log_2(x + 1) = 3$

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(d) $2\log_{10} x + 3 = \log_{10} x^5$

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Question 4 (2 marks)

Solve the following for x :

2

$$81^{x^2} = 3^{-4x-1}.$$

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Question 5 (2 marks)

Phillip has placed \$2 000 000 in an interest-bearing account. He invested at a rate of 5% p.a. Using the compound interest formula shown below, determine the number of years (n) it will take for Phillip to at least double his money.

2

$$A = P(1 + r)^n$$

A (**Amount**) = *Principal plus interest*

P = *Principal*

r = *Interest rate as a percentage*

n = *time in years*

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Question 6 (5 marks)

A logarithmic scale is used to measure the sound level with units in decibels.

$l = 10 \log_{10} \frac{I}{I_o}$, where l is in decibels, I is the intensity of the sound in (watts/m²) and I_o is the threshold of sound intensity detected by the human ear.

- (a) What is the decibel rating of sound at the threshold of hearing (i.e. $I = I_o$). **1**

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- (b) Find the sound level of a conversation, which is 1 million times more intense than the threshold of human hearing. **1**

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- (c) How much greater is the noise intensity of an iPod playing at a peak volume of 115dB (I_1) compared with a half volume level of 85 dB (I_2)? **3**

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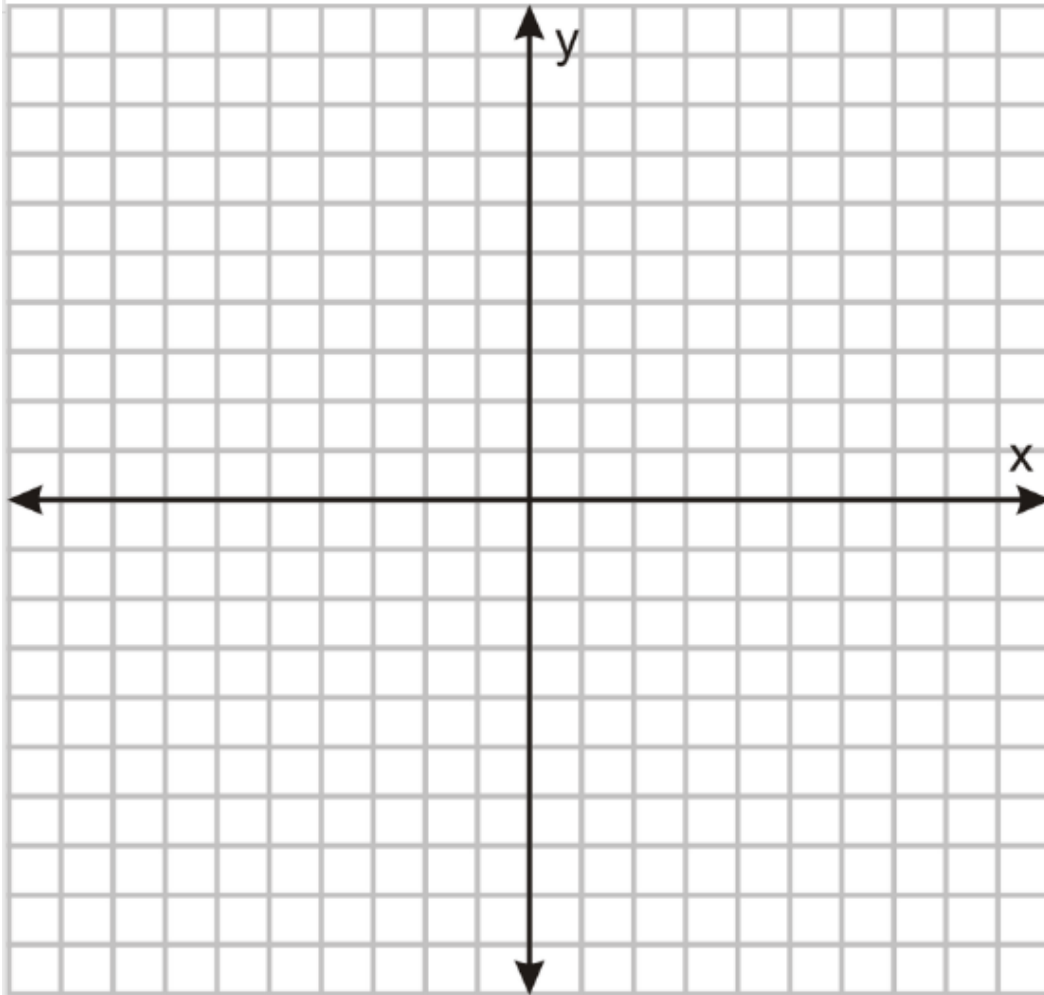
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Question 7 (2 marks)

Sketch the graph of the following exponential function in the box below. Label any intercepts and the asymptote. 2

$$y = 3e^{-x} + 1$$



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Question 8 (6 marks)

Differentiate the following, simplifying where necessary.

(a) $y = (x^2 + x + 1)e^{2x}$

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(b) $y = e^{x - \frac{1}{x}}$

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(c) $y = \frac{3 + e^x}{e^{4x}}$

2

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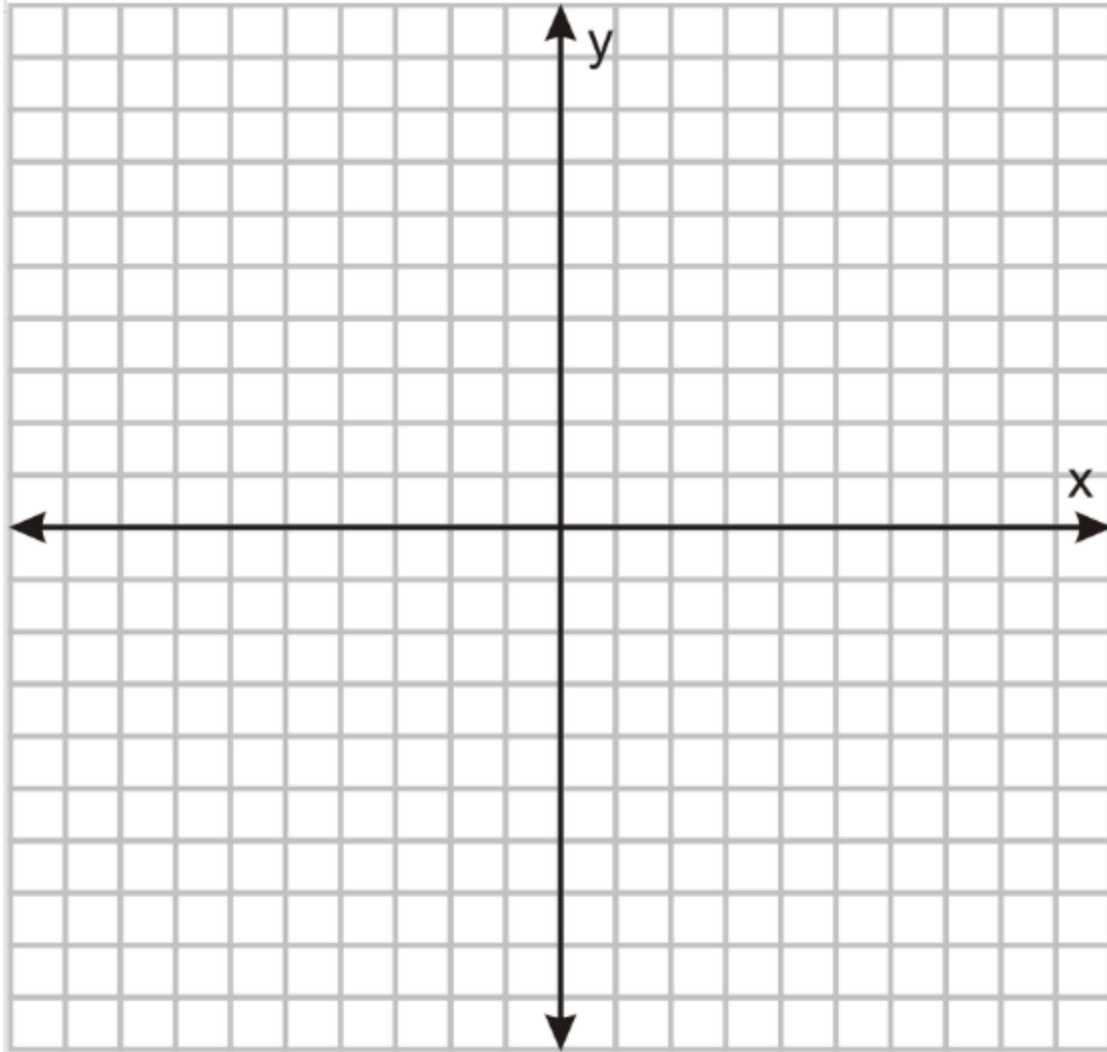
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Question 9 (3 marks)

Sketch the following graph, clearly indicating the vertical asymptote and labelling all critical points. **3**

$$-y = \log_e(x - 2) + 1$$



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END OF EXAMINATION

Examination continues on next page

Extra writing space

If you use this space, clearly indicate which question you are answering.

This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

1.	Solutions and Marking Schemes	Markers Feedback
1(a)	Question 1 (2 marks) If $\log_{10} 2 = 0.3010$ and $\log_{10} 3 = 0.4771$, find: (a) $\log_{10} 6$ 1 $= \log_{10} 2 + \log_{10} 3$ $= 0.3010 + 0.4771$ $= 0.7781$ ① - mark for correct answer with working	
1(b)	(b) $\log_{10} \frac{2}{9}$ 1 $= \log_{10} 2 - \log_{10} 9$ $= 0.3010 - 2 \times 0.4771$ $= -0.6532$ ① - mark for correct answer with working	Some confusion with power rule.
2	Question 2 (2 marks) Solve the following using change of base. Write your answer to three significant figures. 2 $5^x = 3$ $x \ln 5 = \ln 3 \rightarrow$ This is not using change of base $x = \frac{\ln 3}{\ln 5} \rightarrow$ They must show this line for one mark. $x = 0.683$ ① mark for correct answer	Must show change of base in your working, as stated in the question.

3(a)	Solve for x if: (a) $121^x = 11$ 1 $11^{2x} = 11^1$ $2x = 1$ $x = \frac{1}{2}$ or $x = \frac{\log_e 11}{\log_e 121}$ $x = 0.5$ ① - mark for answer.	Some put $x = 2$. Double check your work.
3(b)	(b) $5^{x+3} = 25$ 1 $5^{x+3} = 5^2$ $x+3 = 2$ $x = -1$ ① - mark for correct answer.	
3(c)	(c) $\log_2(x+1) = 3$ 2 $x+1 = 2^3$ $x+1 = 8$ $x = 7$ - ① mark for w/o. - ① mark for correct answer (ECF applies.)	
3(d)	(d) $2\log_{10} x + 3 = \log_{10} x^5$ 2 $2\log_{10} x + 3 = 5\log_{10} x$ $3 = 3\log_{10} x$ $\log_{10} x = 1$ $x = 10$ - ① mark for w/o - ① mark for correct answer.	Some added brackets which changed the question to $2(x+3) = x^5$

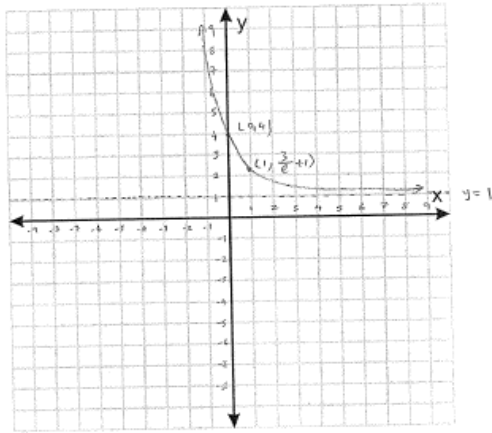
4	Solve the following for x: 2 $81x^2 = 3^{-4x-1}$ $3^{4x^2} = 3^{-4x-1}$ $4x^2 = -4x - 1$ $4x^2 + 4x + 1 = 0 \rightarrow \textcircled{1} - \text{mark}$ $(2x+1)^2 = 0$ $\therefore x = -\frac{1}{2} \quad \textcircled{1} - \text{mark for correct answer}$	Some were stuck on factorising the non-monic quadratic.
5	Phillip has placed \$2 000 000 in an interest-bearing account. He invested at a rate of 5% p.a. 2 Using the compound interest formula shown below, determine the number of years (n) it will take for Phillip to at least double his money. $A = P(1+r)^n$ A (Amount) = Principal plus interest P = Principal r = Interest rate as a percentage n = time in years $4\,000\,000 = 2\,000\,000 (1+5\%)^n$ $2 = 1.05^n$ $\log 2 = n \log 1.05$ $n = \frac{\log 2}{\log 1.05}$ $n = 14.20669985 \dots \rightarrow \textcircled{1} \text{ mark for m.f.o.}$ $n = 15 \text{ years} \rightarrow \textcircled{1} \text{ mark for rounding to the next year.}$	Poorly done. - Many assumed that $5\% = 5$ - Many did not know how to set up the initial equation - Many did not know to round up to an integer, as interest is only compounded at the end of a year.

6(a)	A logarithmic scale is used to measure the sound level with units in decibels. $l = 10 \log_{10} \frac{I}{I_0}$ where l is in decibels, I is the intensity of the sound in (watts/m²) and I_0 is the threshold of sound intensity detected by the human ear. (a) What is the decibel rating of sound at the threshold of hearing (i.e. $I = I_0$). 1 at $I = I_0$ $10 \log_{10} \frac{I_0}{I_0}$ $= 10 \log_{10} 1$ $= 10 \times 0$ $= 0$ — (1) mark w/o ...	
6(b)	(b) Find the sound level of a conversation, which is 1 million times more intense than the threshold of human hearing. 1 $I = 10^6 \cdot I_0$ $10 \log_{10} \left(\frac{10^6 \cdot I_0}{I_0} \right) = 2$ $= 60 \text{ dB}$ (1) mark for substituting $I = I_0$ and an answer of 60 dB	
6(c)	(c) How much greater is the noise intensity of an iPod playing at a peak volume of 115dB (I_1) compared with a half volume level of 85 dB (I_2)? 3 $L_1 = 115 = 10 \log_{10} \left(\frac{I_1}{I_0} \right)$ $L_2 = 85 = 10 \log_{10} \left(\frac{I_2}{I_0} \right)$ $11.5 = \log_{10} \left(\frac{I_1}{I_0} \right)$ $8.5 = \log_{10} \left(\frac{I_2}{I_0} \right)$ $\frac{10}{10} 11.5 = \frac{I_1}{I_0}$ $\frac{10}{10} 8.5 = \frac{I_2}{I_0}$ $I_0 = \frac{I_1}{10^{11.5}}$ (1) mark $I_0 = \frac{I_2}{10^{8.5}}$ (1) mark $\frac{I_2}{I_1} = \frac{10^{8.5}}{10^{11.5}} = 10^{-3} = 0.001$ (1) mark answer: 1000 times greater How much greater is the noise intensity of an iPod playing at a peak volume of 115dB (I_1) compared with a half volume level of 85 dB (I_2)? L	

7

Sketch the graph of the following exponential function in the box below. Label any intercepts and the asymptote. 2

$$y = 3e^{-x} + 1$$



There are no x intercepts.

① - mark for labelling the y intercept

① - mark for a neat sketch

8(a)

Differentiate the following, simplifying where necessary.

$$(a) \quad y = (x^2 + x + 1)e^{2x}$$

$$\frac{dy}{dx} = u'v + uv'$$

$$= e^{2x}(2x+1) + (x^2+x+1) \cdot 2e^{2x}$$

$$= (2x^2 + 2x + 2)e^{2x} + (2x^2 + x)e^{2x}$$

$$= e^{2x}(2x^2 + 4x + 3) \quad \text{--- ① mark for simplifying correctly.}$$

u/v.

$$u = x^2 + x + 1$$

$$u' = 2x + 1$$

$$v = e^{2x}$$

$$v' = 2e^{2x}$$

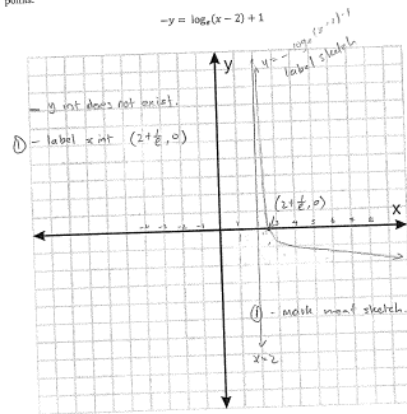
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① - mark for u/v.

8(b)	(b) $y = e^{x-\frac{1}{x}}$ let $u = x - \frac{1}{x}$ 2 $\frac{dy}{du} = e^{u-\frac{1}{u}} \cdot (1 + \frac{1}{u^2})$ $\frac{du}{dx} = 1 + \frac{1}{x^2}$ $\frac{dy}{dx} = (1 + \frac{1}{x^2}) e^{x-\frac{1}{x}}$ $y = e^u$ $\frac{dy}{dx} = e^u \cdot (1 + \frac{1}{x^2})$ $\frac{dy}{dx} = e^{x-\frac{1}{x}} \cdot (1 + \frac{1}{x^2})$ ① ans 1	
8(c)	(c) $y = \frac{3+e^x}{e^{4x}}$ $\frac{u}{v}$ 2 $u = 3 + e^x$ $u' = e^x$ $v = e^{4x}$ $v' = 4e^{4x}$ $\frac{dy}{dx} = \frac{vu' - uv'}{v^2}$ $= \frac{e^{4x} \cdot e^x - (3+e^x) \cdot (4e^{4x})}{(e^{4x})^2}$ $= \frac{e^{5x} - 12e^{4x} - 4e^{5x}}{e^{8x}}$ $= \frac{-11e^{5x} - 12e^{4x}}{e^{8x}}$ — ① correct numerator $= -\frac{11e^{5x}}{e^{8x}} - \frac{12e^{4x}}{e^{8x}}$ — ① correct denominator or $y = 3e^{-4x} + e^{-3x}$ $y' = -12e^{-4x} - 3e^{-3x}$ $= -\frac{12}{e^{4x}} - \frac{3}{e^{3x}} - 8 -$	

9

Sketch the following graph, clearly indicating the vertical asymptote and labelling all critical points.



$$y = -\log_2(x-2) - 1$$